

Bernoulli Sequence

A **Bernoulli sequence** is the name used to describe a sequence of repeated trials with the following properties:

- Each trial results in one of two outcomes, which are usually designated as either a success, S , or a failure, F .
- The probability of success on a single trial, p , is constant for all trials (and thus the probability of failure on a single trial is $1 - p$).
- The trials are independent (so that the outcome of any trial is not affected by the outcome of any previous trial).

Example 1

Suppose that a netball player has a probability of $\frac{1}{3}$ of scoring a goal each time she attempts to goal. She repeatedly has shots for goal. Is this a Bernoulli sequence?

Solution

In this example:

- Each trial results in one of two outcomes, goal or miss.
- The probability of scoring a goal ($\frac{1}{3}$) is constant for all attempts, as is the probability of a miss ($\frac{2}{3}$).
- The trials are independent (so that the outcome of any trial is not affected by the outcome of any previous trial).

Thus, the player's shots at goal can be considered a Bernoulli sequence.

► Bernoulli random variables

The outcome from a Bernoulli trial is represented by a **Bernoulli random variable**, which is a random variable that takes only the values 1 (indicating a success) and 0 (indicating a failure). Thus a Bernoulli random variable Y has a probability distribution of the form:

y	0	1
$\Pr(Y = y)$	$1 - p$	p

In Example 1, each shot at goal can be modelled by a Bernoulli random variable with $p = \frac{1}{3}$, where 1 represents a goal and 0 represents a miss.

► The binomial probability distribution

We can think of a Bernoulli random variable as counting the number of successes in a single Bernoulli trial. What happens if we count the number of successes in a Bernoulli sequence?

The number of successes in a Bernoulli sequence of n trials is called a **binomial random variable** and is said to have a **binomial probability distribution**.

For example, consider rolling a fair six-sided die three times. Let the random variable X be the number of 3s observed. Let T represent a 3, and let N represent not a 3. Each roll meets the conditions of a Bernoulli trial. Thus X is a binomial random variable.

Now consider all the possible outcomes from the three rolls and their probabilities.

Outcome	Number of 3s	Probability	
TTT	$X = 3$	$\frac{1}{6} \times \frac{1}{6} \times \frac{1}{6}$	$\Pr(X = 3) = \left(\frac{1}{6}\right)^3$
TTN	$X = 2$	$\frac{1}{6} \times \frac{1}{6} \times \frac{5}{6}$	$\Pr(X = 2) = 3 \times \left(\frac{1}{6}\right)^2 \times \frac{5}{6}$
TNT	$X = 2$	$\frac{1}{6} \times \frac{5}{6} \times \frac{1}{6}$	
NTT	$X = 2$	$\frac{5}{6} \times \frac{1}{6} \times \frac{1}{6}$	
TNN	$X = 1$	$\frac{1}{6} \times \frac{5}{6} \times \frac{5}{6}$	$\Pr(X = 1) = 3 \times \frac{1}{6} \times \left(\frac{5}{6}\right)^2$
NTN	$X = 1$	$\frac{5}{6} \times \frac{1}{6} \times \frac{5}{6}$	
NNT	$X = 1$	$\frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}$	
NNN	$X = 0$	$\frac{5}{6} \times \frac{5}{6} \times \frac{5}{6}$	$\Pr(X = 0) = \left(\frac{5}{6}\right)^3$

Thus the probability distribution of X is given by the following table.

x	0	1	2	3
$\Pr(X = x)$	$\frac{125}{216}$	$\frac{75}{216}$	$\frac{15}{216}$	$\frac{1}{216}$

Consider the probability that $X = 1$, that is, when exactly one 3 is observed. We can see from the table that there are three ways this can occur. Since the 3 could occur on the first, second or third roll of the die, we can consider this as selecting one object from a group of three, which can be done in $\binom{3}{1}$ ways.

Consider the probability that $X = 2$, that is, when exactly two 3s are observed. Again from the table there are three ways this can occur. Since the two 3s could occur on any two of the three rolls of the die, we can consider this as selecting two objects from a group of three, which can be done in $\binom{3}{2}$ ways.

This leads us to a general formula for this probability distribution:

$$\Pr(X = x) = \binom{3}{x} \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{3-x} \quad x = 0, 1, 2, 3$$

This is an example of the binomial distribution.

If the random variable X is the number of successes in n independent trials, each with probability of success p , then X has a **binomial distribution** and the rule is

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

where $\binom{n}{x} = \frac{n!}{x!(n-x)!}$

Example 2

$n = 7$

Find the probability of obtaining exactly three heads when a fair coin is tossed seven times, correct to four decimal places.

Obtaining a head is considered success here, and the probability of success on each of the seven independent trials is 0.5

Let X be the number of head obtained.

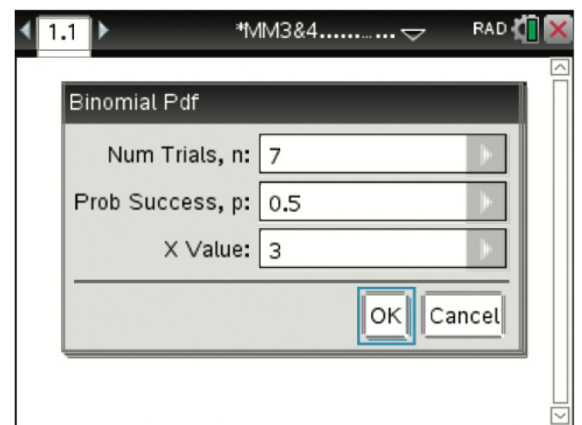
$$\begin{aligned} \Pr(X = 3) &= \binom{7}{3} (0.5)^3 (1 - 0.5)^{7-3} \\ &= 0.2734 \end{aligned}$$

Using the TI-Nspire CX non-CAS

Use **menu** > **Probability** > **Distributions** > **Binomial Pdf** and complete as shown.

Use **tab** or **▼** to move between cells.

The result is shown below.



Note: You can also type in the command and the parameter values directly if preferred.

Example 3

The probability that a person currently in prison has ever been imprisoned before is 0.72. Find the probability that of five prisoners chosen at random at least three have been imprisoned before, correct to four decimal places.

Let X is the number of prisoners who have been imprisoned before,

$$Pr(X = x) = \binom{5}{x} (0.72)^x (1 - 0.72)^{5-x}$$

$$\begin{aligned} \text{Method 1. } Pr(X \geq 3) &= Pr(X=3) + Pr(X=4) + Pr(X=5) \\ &= \binom{5}{3} 0.72^3 (1-0.72)^2 \\ &\quad + \binom{5}{4} 0.72^4 (1-0.72)^1 \\ &\quad + \binom{5}{5} 0.72^5 (1-0.72)^0 \\ &= 0.8624 \end{aligned}$$

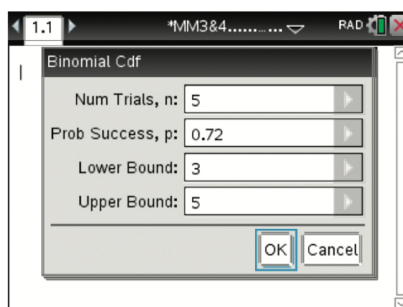
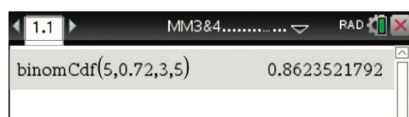
$$\begin{aligned} \text{Method 2. } Pr(X \geq 3) &= 1 - Pr(X=0) - Pr(X=1) \\ &\quad - Pr(X=2) \\ &= 0.8624 \end{aligned}$$

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Use **menu** > **Probability** > **Distributions** > **Binomial Cdf** and complete as shown.

Use **tab** or **▼** to move between cells.

The result is shown below.



Note: You can also type in the command and the parameter values directly if preferred.

Example 4

The probability of a netballer scoring a goal is 0.3. Find the probability that out of six attempts the netballer scores a goal:

- a** four times **b** ~~four times, given that she scores at least one goal.~~

Let X be the number of goals scored.

$$n = 6, \quad p = 0.3$$

$$\begin{aligned}\Pr(X = 4) &= \binom{6}{4} (0.3)^4 (1 - 0.3)^{6-4} \\ &= 0.059535\end{aligned}$$

Section summary

- A **Bernoulli sequence** is a sequence of trials with the following properties:
 - Each trial results in one of two outcomes, which are usually designated as either a success, S , or a failure, F .
 - The probability of success on a single trial, p , is constant for all trials (and thus the probability of failure on a single trial is $1 - p$).
 - The trials are independent (so that the outcome of any trial is not affected by the outcome of any previous trial).
- A **Bernoulli random variable** describes the outcome from a Bernoulli trial; it has a probability distribution of the form $\Pr(Y = 1) = p$ and $\Pr(Y = 0) = 1 - p$.
- The number of successes, X , in a Bernoulli sequence of n trials is called a **binomial random variable** and has a **binomial probability distribution**:

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

$$\text{where } \binom{n}{x} = \frac{n!}{x! (n - x)!}$$

- 2 Find the probability of obtaining exactly four heads when a fair coin is tossed seven times, correct to four decimal places.

$$n = 7 \quad p = 0.5$$

Let X be the number of heads obtained

$$Pr(X=4) = \binom{7}{4} (0.5)^4 (1-0.5)^{7-4} = 0.2734$$

- 3 For a binomial distribution with $n = 4$ and $p = 0.2$, find the probability of:

a three successes

b four successes.

$$\begin{aligned} \text{(a)} \quad Pr(X=3) &= \binom{4}{3} (0.2)^3 (1-0.2)^{4-3} \\ &= 0.0256 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad Pr(X=4) &= \binom{4}{4} (0.2)^4 (1-0.2)^{4-4} \\ &= 0.0016 \end{aligned}$$

- 4 For a binomial distribution with $n = 5$ and $p = 0.4$, find the probability of:

a no successes

b three successes

c five successes.

$$\text{(a)} \quad Pr(X=0) = \binom{5}{0} (0.4)^0 (1-0.4)^{5-0} = 0.0778$$

$$\text{(b)} \quad Pr(X=3) = \binom{5}{3} (0.4)^3 (1-0.4)^{5-3} = 0.2304$$

$$\text{(c)} \quad Pr(X=5) = \binom{5}{5} (0.4)^5 (1-0.4)^{5-5} = 0.01024$$

7 A fair die is rolled six times and the number of 2s noted. Find the probability of:

a exactly three 2s

b more than three 2s

c at least three 2s.

$$n = 6 \quad p = \frac{1}{6}$$

Let x be the number of 2s obtained.

$$(a) \Pr(X=3) = \binom{6}{3} \left(\frac{1}{6}\right)^3 \left(1 - \frac{1}{6}\right)^{6-3} = 0.0536$$

$$\begin{aligned}(b) \Pr(X > 3) &= \Pr(X=4) + \Pr(X=5) + \Pr(X=6) \\&= \binom{6}{4} \left(\frac{1}{6}\right)^4 \left(1 - \frac{1}{6}\right)^{6-4} + \binom{6}{5} \left(\frac{1}{6}\right)^5 \left(1 - \frac{1}{6}\right)^{6-5} \\&\quad + \binom{6}{6} \left(\frac{1}{6}\right)^6 \left(1 - \frac{1}{6}\right)^{6-6} \\&= 0.0087\end{aligned}$$

$$\begin{aligned}(c) \Pr(X \geq 3) &= \Pr(X=3) + \Pr(X=4) + \Pr(X=5) + \Pr(X=6) \\&= \binom{6}{4} \left(\frac{1}{6}\right)^4 \left(1 - \frac{1}{6}\right)^{6-4} + \binom{6}{5} \left(\frac{1}{6}\right)^5 \left(1 - \frac{1}{6}\right)^{6-5} \\&\quad + \binom{6}{6} \left(\frac{1}{6}\right)^6 \left(1 - \frac{1}{6}\right)^{6-6} + \binom{6}{3} \left(\frac{1}{6}\right)^3 \left(1 - \frac{1}{6}\right)^{6-3} \\&= 0.0623\end{aligned}$$

- 8 Jo knows that each ticket has a probability of 0.1 of winning a prize in a lucky ticket competition. Suppose she buys 10 tickets.
- a Write down a general rule for the probability distribution of the number of winning tickets.
- b Use the rule to calculate the probability that Jo has:
- i no wins ii at least one win.

(a) $n = 10$ $p = 0.1$

Let X be the number of winning tickets.

$$Pr(X = x) = \binom{10}{x} (0.1)^x (1 - 0.1)^{10 - x}$$

$$x = 0, 1, \dots, 10$$

(b) i) $Pr(X = 0) = \binom{10}{0} (0.1)^0 (1 - 0.1)^{10 - 0}$
 $= 0.3487$

ii) $Pr(X \geq 1) = 1 - Pr(X = 0)$
 $= 1 - 0.3487$
 $= 0.6513$

- 9 Suppose that the probability that a person selected at random is left-handed is always 0.2. If 11 people are selected at random for the cricket team:
- a Write down a general rule for the probability distribution of the number left-handed people on the team.
 - b Use the rule to calculate the probability of selecting:
 - i exactly two left-handers
 - ii no left-handers
 - iii at least one left-hander.

$$n = 11 \quad p = 0.2$$

$$(a) \quad \Pr(X = x) = \binom{11}{x} (0.2)^x (0.8)^{11-x}$$
$$x = 0, 1, \dots, 11$$

$$(b) \quad (i) \quad \Pr(X = 2) = \binom{11}{2} 0.2^2 0.8^9 = 0.2953$$

$$(ii) \quad \Pr(X = 0) = \binom{11}{0} 0.2^0 0.8^{11} = 0.0859$$

$$(iii) \quad \Pr(X \geq 1) = 1 - \Pr(X = 0)$$
$$= 0.9141$$

- 11** The probability of a particular drug causing side effects in a person is 0.2. What is the probability that at least two people in a random sample of 10 people will experience side effects?

$$n = 10 \quad p = 0.2$$

$$\begin{aligned} P_r(X \geq 2) &= 1 - P_r(X=0) - P_r(X=1) \\ &= 1 - \binom{10}{0} (0.2)^0 (1-0.2)^{10-0} \\ &\quad - \binom{10}{1} (0.2)^1 (1-0.2)^{10-1} \\ &\approx 0.624 \end{aligned}$$

- 20** A multiple-choice test has eight questions, each with five possible answers, only one of which is correct. Find the probability that a student who guesses the answer to every question will have:

- a** no correct answers
- b** six or more correct answers

$$n = 8 \quad p = 0.2$$

$$\begin{aligned} \text{(a)} \quad P_r(X=0) &= \binom{8}{0} (0.2)^0 (1-0.2)^{8-0} \\ &= 0.1678 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P_r(X \geq 6) &= P_r(X=6) + P_r(X=7) + P_r(X=8) \\ &= 0.00123 \end{aligned}$$

$$\begin{aligned} &\binom{8}{6} (0.2)^6 (1-0.2)^{8-6} + \binom{8}{7} (0.2)^7 (1-0.2)^{8-7} \\ &\quad + \binom{8}{8} (0.2)^8 (1-0.2)^{8-8} \end{aligned}$$

QUESTION 8

$$n = 6 \quad p = 0.25$$

A test includes six multiple choice questions. Each question has four options for the answer.

If the answers are guessed, the probability of getting at most two questions correct is represented by

(A) $\binom{6}{0} 0.25^0 \times 0.75^6 + \binom{6}{1} 0.25^1 \times 0.75^5$

$$Pr(X \leq 2)$$

$$Pr(X=0) + Pr(X=1) + Pr(X=2)$$

(B) $\binom{6}{0} 0.25^0 \times 0.75^6 + \binom{6}{1} 0.25^1 \times 0.75^5 + \binom{6}{2} 0.25^2 \times 0.75^4$

(C) $1 - \left(\binom{6}{0} 0.25^0 \times 0.75^6 + \binom{6}{1} 0.25^1 \times 0.75^5 \right)$

(D) $1 - \left(\binom{6}{0} 0.25^0 \times 0.75^6 + \binom{6}{1} 0.25^1 \times 0.75^5 + \binom{6}{2} 0.25^2 \times 0.75^4 \right)$

QUESTION 6

A random variable X is the number of successes in a Bernoulli experiment with n trials, each with a probability of success p and a probability of failure q . The probability distribution table of X is shown.

$$n = 4$$

$$Pr(X=0)$$

$$= \binom{4}{0} p^0 q^4 = \frac{1}{81}$$

$$q = \frac{1}{3}$$

k	$P(X=k)$
0	$\frac{1}{81}$
1	$\frac{8}{81}$
2	$\frac{24}{81}$
3	$\frac{32}{81}$
4	$\frac{16}{81}$

$$\binom{n}{n} = 1$$

$$\binom{n}{0} = 1$$

$$n = 4$$

Which values of n , p and q will generate this probability distribution?

(A) $n = 4, p = \frac{2}{3}, q = \frac{1}{3}$

(B) $n = 4, p = \frac{1}{3}, q = \frac{2}{3}$

(C) $n = 5, p = \frac{2}{3}, q = \frac{1}{3}$

(D) $n = 5, p = \frac{1}{3}, q = \frac{2}{3}$